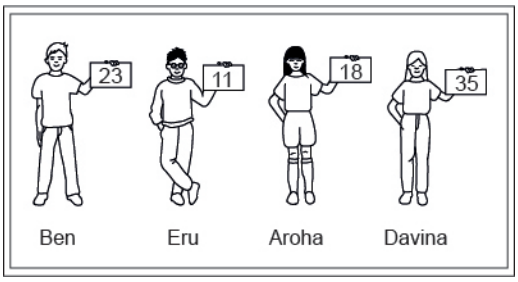
Choose a circle to show how much each sentence is like you	Very Unlike Me 1	Unlike Me 2	Like Me 3	Very Like Me 4
01. I like maths at school.	⬭	⬭	⬭	⬭
02. I am good at maths.	⬭	⬭	⬭	⬭
03. My teacher thinks I am good at maths.	⬭	⬭	⬭	⬭
04. My family/whānau think I am good at maths.	⬭	⬭	⬭	⬭
05. I enjoy doing maths in my own time (not at school).	⬭	⬭	⬭	⬭
06. I enjoy doing things in maths that I haven't tried before.	⬭	⬭	⬭	⬭

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Practice Questions

These practice questions are to help you understand how to show your answer for different types of questions.

P01. Who is holding a card with an even number on it?

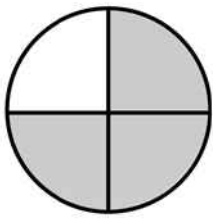


- ⬭ Ben
- ⬭ Eru
- ⬭ Aroha
- ⬭ Davina

P02. Complete this number pattern.

2, 4, ____, ____, 10

P03. What fraction of this circle is shaded?



☐

———

☐

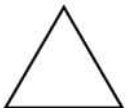
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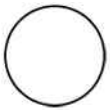
P04. Match the sentence with the correct shape.

☐ 1. I have three sides

☐ 2. I have 4 sides

a. 

b. 

c. 

P05. Which numbers make this number sentence **TRUE**?



☐ 1

☐ 2

☐ 3

☐ 4

☐ 5

P06. Put the numbers 1, 2, 3, and 4 in the boxes to order these numbers from biggest (1) to smallest (4).

☐ 3

☐ 7

☐ 2

☐ 0

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P07. Select whether the following statements are True or False.

	TRUE	FALSE
In the number 213, the value of 1 is ten.	⬭	⬭
In the number 504, the value of 5 is fifty.	⬭	⬭

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01. Mason has these shirts and shorts to choose from shown below.
Which shows all the different ways he can combine 1 shirt and 1 pair of shorts?



⬭ 

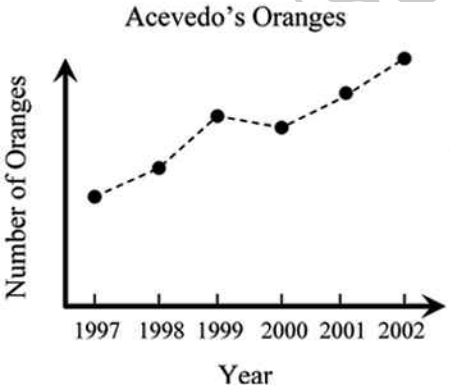
⬭ 

⬭ 

⬭ 

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02. The graph below shows the number of oranges harvested from the Acevedo's orange tree over a 6-year period.
Which of the following is a correct conclusion about the graph?



⬭ Fewer oranges were harvested in 2002 than in 1997.

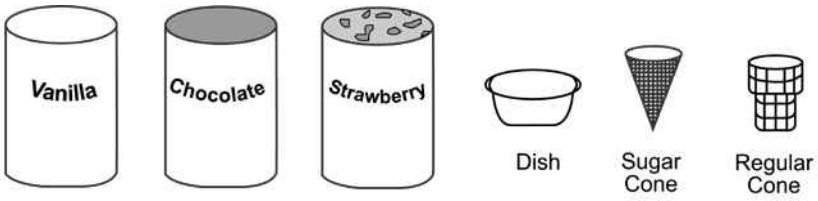
⬭ More oranges were harvested in 2001 than in 1998.

⬭ The number of oranges harvested increased every year.

⬭ The number of oranges harvested decreased from 2000 to 2001.

Use the following information to answer question03..

Jan's Snack Shop has 3 flavours of ice cream - vanilla, chocolate, and strawberry.
The ice cream can be served in a dish, a sugar cone, or a regular cone.
There are 9 people who choose 1 dip of ice cream in a dish, or in a sugar cone, or in a regular cone, and all of their choices are different.



03. List the 9 different choices.

_____

_____

End of Section

04. In a fruit bowl there were 5 bananas, 6 apples, 4 oranges and 8 plums. Joanna wants a piece of fruit. She doesn't care what kind she gets.



What chance does she have of picking an orange?

- ⬭ Poor
- ⬭ Good
- ⬭ Certain
- ⬭ Even

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05. Every Sunday in autumn, Martin has to do 1 outside job and 1 inside job. The jobs are listed below.
How many different combinations of 1 inside job and 1 outside job can Martin make?

Inside Jobs	Outside Jobs
vacuum	rake
wash dishes	weed
dust	

- ⬭ 3
- ⬭ 5
- ⬭ 6
- ⬭ 9

06. There are three Year 8 students and two Year 7 students on the swim team.
Everyone's name is put in a hat and the captain is chosen by picking one name.
What are the chances that the captain will be a Year 8 student?

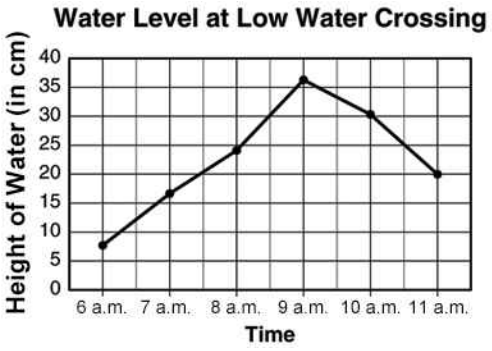
- ⬭ 1 out of 5
- ⬭ 1 out of 3
- ⬭ 3 out of 5
- ⬭ 2 out of 3

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07. In Sheila's class, there are 5 students with red hair, 8 students with blond hair, and 9 students with brown hair.
Which of the following questions about these students could you use probability to solve?

- ⬭ How many students are in Shelia's class altogether?
- ⬭ If the teacher picks 1 student at random, what colour hair is that student most likely to have?
- ⬭ How many more students have brown hair than red hair?
- ⬭ If the teacher buys 2 pencils for each student, how many pencils will she buy?

08. After a heavy rain storm, Mrs Mendez recorded the height of the water at a low water crossing. She used her data to make the graph below.
Which is ***closest*** to the height of the water recorded at 6:30 am?



⬭ 8 cm

⬭ 10 cm

⬭ 13 cm

⬭ 15 cm

09. The students in Ms Romero's social studies class are preparing to learn about South American countries. The table below shows possible report topics.
Each student will select a country, a geographic feature to study, and a visual display.
How many different types of reports with one country, one feature, and one display can the students write?

Country	Geographic Feature	Visual Display
Columbia	Mountain	Map
Chile	River	Flag
Argentina	Lake	Currency
Brazil		

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⬭ 80

⬭ 36

⬭ 10

⬭ 4

10. This is a list of Beth's English homework scores for the grading period.
93, 83, 64, 84, 76, 83, 78, 76, 60, 81
Which stem-and-leaf plot correctly displays the information?

Stem	Leaf
6	II
7	III
8	IIII
9	I

Stem	Leaf
6	4
7	6,8
8	1, 3, 4
9	3

Stem	Leaf
6	4
7	6,6, 8
8	1, 3 3, 4
9	3

Stem	Leaf
6	0, 4
7	6, 6, 8
8	1, 3, 3, 4
9	3

ADMINISTER ONSCREEN ONLY

11. This is a stem-and-leaf plot of a group of test scores.
What is the median score?

Stem	Leaf
5	3 4
6	2 4 8
7	0 1 2 5 7 7 9
8	4 5 6 7
9	1 2 4 6

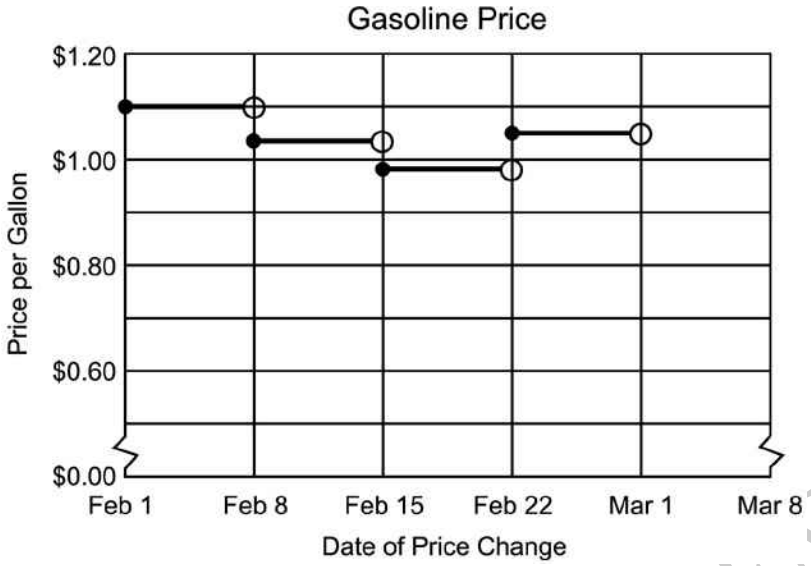
- ⬭ 73
- ⬭ 76
- ⬭ 77
- ⬭ 77.5

12. Chad has a mean time for running the under 200-metre dash of 36.2 seconds.
Which of the following statements **MUST** be **TRUE**?

- ⬭ Chad runs the 200-metre dash in 0 to 36.2 seconds.
- ⬭ Chad's best time in the 200-metre dash is 36.2 seconds.
- ⬭ Chad runs the 200-metre dash, on average, in 36.2 seconds.
- ⬭ Chad runs the 200-metre dash faster than 36.2 seconds most of the time.

ADMINISTER ON SCREEN ONLY

13. Mr Mitchell owns a petrol station. He changes the prices of petrol once each week. The graph below shows the weekly prices in February.
Which of the following statements is supported by the data on the graph?



- ⬭ The price of petrol rose each week.
- ⬭ Petrol was cheapest the week of February 1.
- ⬭ The average price in February was less than $1.00.
- ⬭ On February 28 the price of petrol was about $1.05.
- ⬭ The price of petrol on March 15 will be $1.10.

ADMINISTER ON SCREEN ONLY

14. A local restaurant is advertising a combination dinner special. Donna can choose one entrée, one side, and one drink.
According to the menu, from how many different dinner combinations can Donna choose?

Entrée	Side	Drink
Hamburger	Salad	Milk
Chicken Sandwich	Fruit	Juice
Lasagna	Chips	Soda

- ⬭ 27
- ⬭ 18
- ⬭ 9
- ⬭ 3

15. Out of 36 students, four students are named John. Each of the students puts his or her name on an index card.
If a card is drawn at random, what is the probability it will say John?

- ⬭ $\frac{1}{36}$
- ⬭ $\frac{1}{12}$
- ⬭ $\frac{1}{9}$
- ⬭ $\frac{1}{4}$

ADMINISTER ONSCREEN ONLY

16. A journalist is interested in estimating the percentage of Northland residents who support the use of tax dollars to pay for a new civic centre in Whangarei.
Which of the following would result in the **MOST** reliable estimate?

- ⬭ Survey 100 randomly selected civic centre board members.
- ⬭ Survey 100 randomly selected Auckland residents.
- ⬭ Survey 100 randomly selected Whangarei residents.
- ⬭ Survey 100 randomly selected Northland residents.

17. A bag contains 8 blue, 3 red, and 6 white chips. Only red chips are added to the bag. How many red chips must be added to the bag for the probability of drawing a red chip to be $\frac{1}{3}$?

- ⬭ 2
- ⬭ 3
- ⬭ 4
- ⬭ 6

Use the following information to answer questions 18 to 20.

Below is a stem-and-leaf graph showing the results of a spelling test.

Only Melanie got them all correct.

3	0												
2	0	1	2	2	2	2	4	4	5	6	6	8	9
1	0	2	4	5	5	6	6	7	7	7	8	9	9
0	8	9											

18. How many students scored less than 12?

_____

19. What mark did Melanie get?

_____

20. Which mark scored **MOST** frequently?

_____

End of Section

ADMINISTER ONLY

21. Patrice and Tom need to report the results of a survey regarding the favourite snack food of the students at Milton High School.

How could the experiment be done to produce a random sampling of 100 students?

- ⬭ Ask the students as they get off the school buses.
- ⬭ Ask the opinion of all the teachers at the school.
- ⬭ Ask all the students who are in the school cafeteria during one day.
- ⬭ Ask every 20th student on the school roll until 100 students have been asked.

22. Dave is a member of a bowling league. The table below shows his record for the last 10 games (100 frames) he bowled.
Based on his previous record, what is the probability that Dave will bowl a strike in the next frame?

Dave's Bowling Results

Type of Ball	Number of Frames
Strike	12
Spare	26
Gutter ball	3
Other	59

- ⬭ $\frac{3}{100}$
- ⬭ $\frac{3}{50}$
- ⬭ $\frac{3}{25}$
- ⬭ $\frac{6}{25}$

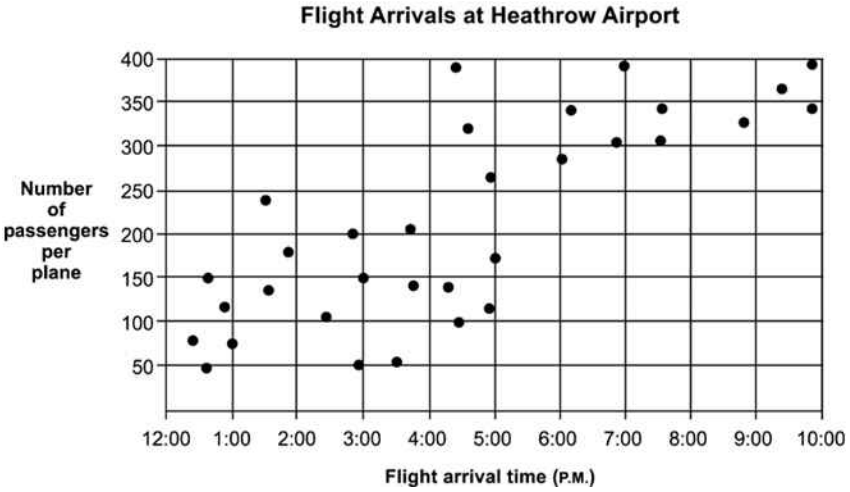
ADMINISTER ONLY

23. A journalist is interested in estimating the percentage of Massachusetts residents who support the use of state tax dollars to pay for a new civic centre in Springfield.
Which of the following would result in the **MOST** reliable estimate?

- ⬭ Survey 100 randomly selected civic centre board members.
- ⬭ Survey 100 randomly selected Boston residents.
- ⬭ Survey 100 randomly selected Springfield residents.
- ⬭ Survey 100 randomly selected Massachusetts residents.

Use the following information to answer question24..

The number of passengers on each plane landing at an airport is recorded and shown below.



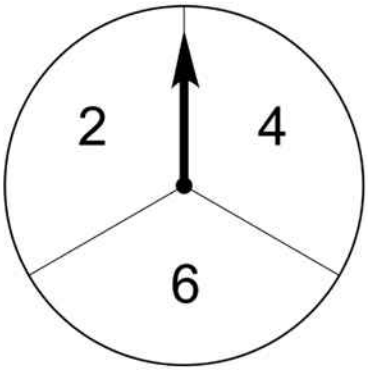
24. The scatter plot shows that

- ⬭ fewer planes land in the afternoon than in the evening.
- ⬭ there are more passengers per plane in the evening than in the afternoon.
- ⬭ more planes land in the evening than in the afternoon.
- ⬭ there are more passengers per plane in the afternoon than in the evening.

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End of Section

25. Ramón is going to spin the arrow on the spinner twice and add the results.
What is the **MOST LIKLEY** sum of the two spins?



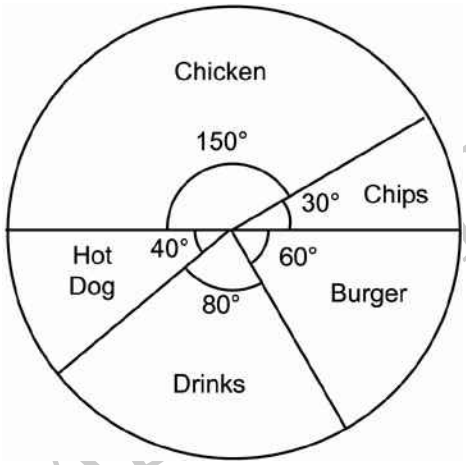
⬭ 6

⬭ 8

⬭ 10

⬭ 12

26. Jenny is keen to try her luck at a fair and pays $2 for the wheel of fortune.



ANSWER ONSCREEN ONLY

What is her probability (p, written as fraction), of winning a chicken?

p =

$$\frac{\square}{\square}$$

27. The table shows the mean number of points scored per game by four professional basketball players in four seasons.
Which player had the greatest range of mean points per game for the seasons shown in the table?

MEAN POINTS SCORED PER GAME

	Season			
Player	1998-1999	1999-2000	2000-2001	2001-2002
Alonzo Mourning	20.1	19.2	19.8	23.2
Tim Hardaway	17.4	18.9	20.3	17.2
Jamal Mashburn	14.8	15.1	13.4	10.6
Terry Mills	9.0	14.2	10.8	9.4

⬭ Alonzo Mourning

⬭ Tim Hardaway

⬭ Jamal Mashburn

⬭ Terry Mills

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